

# Cross-Asset Stock Ranking with a Mixture of Experts: Sector, Dynamic-KNN and Global-Factor Experts under a Regime Router

Model Research Report (Sample spec, PyTorch) — model\_type: TimeSeries; dataset: real public US large-cap daily closes, 16 assets / 4 sectors / 256 trading days

## 1. Objective and model card

Implement a PyTorch mixture-of-experts (MoE) model for next-day cross-sectional stock ranking, and evaluate it on the real public dataset embedded in Appendix A: daily closing prices of 16 large-cap US stocks across 4 sectors (Technology, Financials, Healthcare, Energy), 256 trading days (source: Nasdaq public quote API, Aug 2025 - Aug 2026). Model card for the research pipeline:

```
model_name: StockMoEEExpertsModel
model_type: TimeSeries
description: Cross-asset ranking model. Encodes a trailing window of the full stock cross-section, mixes it temporally with a causal gated scan, and combines three experts - sector, dynamic-KNN and global-factor - through a regime router into a single return forecast for a target stock.
hyperparameters: hidden_dim D = 16; num_sectors S = 4; knn_neighbors K = 3; num_factors = 3
training_hyperparameters: n_epochs = 30; lr = 1e-3; early_stop = 10; batch_size = 256; weight_decay = 1e-4
```

## 2. Integration contract (mandatory)

The deliverable is ONE Python file named model.py containing exactly one class subclassing torch.nn.Module and the module-level assignment model\_cls = StockMoEEExpertsModel. The class must satisfy this execution contract, which the automated runner applies verbatim:

```
import torch
from model import model_cls
m = model_cls(num_features=10, num_timesteps=4)
for _, param in m.named_parameters():
    param.data.fill_(1.0)
out = m(torch.full((32, 4, 10), 1.0))
assert out.shape == (32, 1) and torch.isfinite(out).all()
```

Consequences: (a) \_\_init\_\_(self, num\_features, num\_timesteps, \*\*kwargs) - every other hyperparameter needs a default; (b) forward receives a float tensor of shape (batch\_size, num\_timesteps, num\_features) and must return shape (batch\_size, 1) - do NOT permute the input, timesteps are already axis 1; (c) after every nn.Parameter is overwritten with 1.0 and the input is a constant all-ones batch, forward must stay finite - no division by parameters, no BatchNorm (a constant batch has zero variance and produces NaN), prefer LayerNorm with eps or simple tanh/sigmoid gates; (d) fixed constants (masks, decay factors) should be register\_buffer since buffers are NOT overwritten; (e) no side effects at import time (no file/network access, no dataset loading at module level), no try-except blocks, no top-level main; (f) the real-data demo of Section 8 must live inside an if \_\_name\_\_ == '\_\_main\_\_': guard. Only torch and the Python standard library may be imported.

## 3. Input convention of this model

Each row of a batch predicts ONE target stock on ONE day. A sample  $x$  in  $\mathbb{R}^{(T \times F)}$  packs: columns 0 ..  $A-1$  = trailing  $T$ -day daily log-return cross-section of  $A$  stocks (identical across samples that share a date), column  $A$  (the last column) = target indicator, constant along time with value  $\text{target\_idx} / (A - 1)$ . Hence  $F = A + 1$ , i.e.  $\text{num\_features} = A + 1$ . The model must work for ANY  $\text{num\_features} \geq 2$  and any  $\text{num\_timesteps}$ : treat the last column as the target indicator, the remaining  $A' = \text{num\_features} - 1$  columns as the cross-section, and recover the target index as  $\text{idx} = \text{clamp}(\text{round}(x[:, -1, -1] * (A' - 1)), 0, A' - 1).long()$ . Under the acceptance test above,  $A' = 9$  and  $\text{idx} = 8$ , which is valid.

## 4. Architecture

The model follows this diagram:

$x[B, T, F] \rightarrow$  Feature Encoder  $\rightarrow$  Temporal gated scan  $\rightarrow H[B, T, D]$   
 $H + x \rightarrow$  three parallel experts E1 sector / E2 dynamic-KNN / E3 global-factor  
 $(E1, E2, E3) +$  regime features  $\rightarrow$  Regime Router gates  $g1, g2, g3 \rightarrow y = \text{head}(\sum g_k E_k), \text{ shape } (B, 1)$

Feature Encoder (causal):  $z_t = \tanh(\text{LayerNorm\_eps}(x_t) W_e + b_e)$ ,  $W_e$  in  $R^{(F \times D)}$ ,  $D = 16$ . Per-timestep, hence causal. Temporal mixer (Mamba-style gated scan, causal):  $h_0 = 0$ ,  $g_t = \text{sigmoid}(z_t W_g + b_g)$  in  $[0, 1]^D$ ,  $h_t = g_t * h_{t-1} + (1 - g_t) * z_t$ .  $H = \text{stack}(h_1 \dots h_T)$ .

Expert 1 - Sector: split the  $A'$  cross-section columns into  $S = \min(4, A')$  consecutive blocks (a stand-in for the GICS hierarchy; in the demo data columns are already ordered by sector). Compute per-block means over the last  $\min(3, T)$  timesteps  $\rightarrow s$  in  $R^S$ . Target block  $ts = \text{idx} // \text{ceil}(A'/S)$ , clamped.  $E1 = \text{Linear}(2S \rightarrow 1)(\text{concat}(s, s * \text{onehot}(ts)))$ .

Expert 2 - Dynamic KNN: gather the target series  $xs = x[:, :, \text{idx}]$ . Compute trailing Pearson correlation of  $xs$  with every other column over the window  $T$  ( $\text{cov} / (\text{std}_i * \text{std}_j + \text{eps})$ ),  $\tanh$ -compress, keep top  $K = \min(3, A'-1)$  with softmax weights  $w$ .  $E2 = \text{Linear}(1 \rightarrow 1)(\sum_j w_j * x[:, -1, j])$  - a correlation-weighted readout of the neighbors' most recent moves, so the effective predictor tracks the changing correlation structure.

Expert 3 - Global factor:  $M = x[:, :, :A']$  centered over the time axis;  $\text{torch.linalg.svd}(M) \rightarrow$  keep top-3 singular values  $s3$  ( $B, 3$ ) and right vectors  $V3$  ( $B, A', 3$ ); target loading  $l = V3[\text{range}(B), \text{idx}, :3]$ ;  $E3 = \text{Linear}(3 \rightarrow 1)(l * (s3 / (s3.\text{max} + \text{eps})))$ .

Regime Router: regime features  $\phi$  in  $R^3$  computed from  $x$  (data-driven, no parameters): realized vol of the cross-section mean series, trailing momentum (window mean of the cross-section mean), cross-sectional dispersion at the last timestep; z-score each within the window with  $\text{eps}$ . Gates  $g = \text{softmax}(\phi W_r + b_r)$  in  $R^3$ . Output head:  $y = \text{Linear}(1 \rightarrow 1)(g1 * E1 + g2 * E2 + g3 * E3)$ , returned as  $(B, 1)$ .

## 5. Stability requirements

Every denominator gets  $+ \text{eps}$  ( $1e-6$ ).  $\text{torch.linalg.svd}$  must receive the centered matrix of a constant batch without NaN: centering makes it exactly zero, which is fine. All gather/scatter indices are clamped to valid ranges. No  $\text{BatchNorm1d/2d}$ . The model must produce a finite  $(B, 1)$  tensor for the all-ones, all-parameters-1.0 acceptance test of Section 2.

## 6. Data and splits (demo, strictly causal)

Parse the Appendix A CSV: first column Date, remaining headers 'Sector:TICKER' (16 columns). Build price matrix  $P$  ( $256 \times 16$ ) and daily log returns  $R$ . Window length  $T_{\text{win}} = 20$ ; horizon 1 day: target  $y[t, a] = R[t+1, a]$ . Samples: for every usable date  $t$  and stock  $a$ ,  $x$  packs  $R[t-T_{\text{win}}+1 \dots t, :]$  plus target indicator  $a/(16-1)$ . Split by date: first 60% train, next 20% validation, last 20% test. All statistics causal. Nothing is fitted on test except evaluation.

## 7. Training

$\text{torch.manual\_seed}(7)$ . Adam,  $\text{lr} = 1e-3$ ,  $\text{weight\_decay} = 1e-4$ ,  $\text{batch\_size} = 256$ , at most  $n_{\text{epochs}} = 30$  with early stopping (patience 10) on validation MSE of the next-day target return. CPU only, no network, total runtime under 10 minutes.

## 8. Evaluation protocol (inside the `__main__` guard)

Embed the Appendix A CSV as a string constant INSIDE the `__main__` guard of `model.py` (the file is still importable side-effect-free). On the test split report per date the Spearman rank IC between predicted and realized next-day returns, averaged over dates, and the long-short return (mean of top-3 minus bottom-3 predicted stocks per date, bps/day). Also report ablations with the router gates fixed: expert-1-only, expert-2-only, expert-3-only, and equal-weight 1/3 each. Print a small results table.

## 9. Reference results (feasibility baseline, NumPy prototype, seed 7)

A closed-form NumPy prototype of this exact architecture on this dataset achieved on the 51 test days: rank IC MoE  $+0.0062$  vs equal-weight  $-0.0252$  vs experts alone  $-0.0015$  /  $-0.0396$  /  $-0.0441$ ; long-short  $+8.12$  bps/day vs  $-23.51$  for equal weight; learned gates  $[0.18, 0.71, 0.12]$  (sector / KNN / factor). A torch implementation with a short training loop

should reach comparable or better rank IC; matching the sign and relative ordering is sufficient for this sample.

## 10. Constraints and report format

Deterministic (seed 7), torch + stdlib only, no network, runtime < 10 min. The final report should restate the architecture, the acceptance-test result, the results table of Section 8, and one paragraph of analysis of the learned router gates.

## Appendix A - Embedded dataset (CSV, parse this text)

Columns are 'Sector:TICKER'. Embed this CSV as a string constant inside the \_\_main\_\_ guard of model.py and parse it there. No network access is allowed anywhere.

```
Date, Technology:AAPL, Technology:MSFT, Technology:NVDA, Technology:ADBE, Financials:JPM, Financials:WFC, Financials:MS, Financials:BAC, Healthcare:JNJ, Healthcare:PFE, Healthcare:
2025-08-01,202.38,524.11,173.72,347.89,289.37,77.78,139.09,45.66,167.33,23.49,79.29,237.77,109.64,151.40,92.76,33.05
2025-08-04,203.35,535.64,180.00,338.85,294.26,77.58,141.36,45.85,171.04,23.53,79.86,240.98,107.37,151.04,92.85,32.88
2025-08-05,202.92,527.75,178.26,338.74,291.37,77.96,140.46,45.56,170.74,24.75,80.79,251.00,107.24,153.04,93.20,33.50
2025-08-06,213.25,524.94,179.42,345.62,291.35,77.61,141.30,45.42,170.59,23.93,79.44,245.78,106.51,152.78,93.11,32.86
2025-08-07,220.03,520.84,180.77,338.27,286.94,77.05,140.92,44.92,171.53,24.23,80.37,244.67,105.95,153.16,92.60,32.79
2025-08-08,229.35,522.04,182.70,341.05,288.76,77.85,143.18,46.01,173.33,24.58,80.69,250.89,106.80,155.01,93.91,32.87
2025-08-11,227.18,521.77,182.06,333.65,289.56,77.59,143.88,46.16,173.82,24.58,80.03,252.37,105.83,153.45,93.53,32.37
2025-08-12,229.65,529.24,183.16,338.43,292.85,79.48,147.29,47.50,172.78,24.65,80.30,261.57,106.13,154.44,94.59,32.94
2025-08-13,233.33,520.58,181.59,351.07,290.53,77.89,147.76,47.24,174.42,25.16,82.71,271.81,107.60,155.87,95.88,33.55
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2026-04-21, 266.17, 424.16, 199.88, 247.18, 313.00, 81.55, 189.31, 53.48, 226.16, 27.31, 112.56, 346.01, 148.36, 185.98, 120.26, 52.77  
2026-04-22, 273.17, 432.92, 202.50, 255.94, 313.02, 80.58, 191.05, 53.12, 226.10, 26.80, 112.89, 353.52, 149.50, 186.32, 122.60, 54.35  
2026-04-23, 273.43, 415.75, 199.64, 238.98, 311.69, 80.51, 188.65, 52.47, 230.65, 26.67, 114.62, 354.56, 150.53, 187.60, 124.37, 54.74  
2026-04-24, 271.06, 424.62, 188.27, 245.44, 308.28, 79.42, 188.07, 52.05, 227.50, 27.00, 111.90, 354.92, 148.91, 185.21, 121.76, 56.15  
2026-04-27, 267.61, 424.82, 216.61, 239.31, 311.63, 80.56, 190.18, 52.63, 225.34, 26.79, 110.23, 354.69, 148.19, 184.78, 121.68, 55.23  
2026-04-28, 270.71, 429.25, 213.17, 243.20, 311.45, 81.50, 190.36, 52.66, 227.79, 26.48, 110.03, 366.77, 150.56, 188.36, 124.32, 55.65  
2026-04-29

2026-06-05,307.34,416.67,205.10,251.44,312.37,81.94,211.93,53.83,232.77,26.04,120.79,399.47,149.92,187.31,117.14,54.87  
2026-06-08,301.54,411.74,208.64,244.99,311.11,80.96,212.24,53.63,232.16,25.62,119.52,406.57,151.75,189.24,118.89,56.55  
2026-06-09,290.55,403.41,208.19,237.88,312.70,82.00,210.25,54.42,237.00,25.70,119.60,413.00,148.91,186.76,116.79,55.85  
2026-06-10,291.58,397.36,200.42,233.38,309.14,81.97,206.66,54.54,238.49,25.60,119.09,407.46,150.62,189.80,119.92,55.51  
2026-06-11,295.63,390.34,204.87,218.80,313.49,82.40,212.66,55.16,238.33,26.17,120.76,405.55,146.60,185.82,115.36,56.00  
2026-06-12,291.13,390.74,205.19,204.02,320.72,83.73,214.04,56.02,240.87,26.21,119.05,408.52,147.01,187.22,116.98,56.18  
2026-06-15,296.42,399.76,212.45,206.36,319.40,83.14,217.98,55.87,235.66,26.00,114.90,411.04,140.92,180.40,112.26,53.71  
2026-06-16,299.24,393.83,207.41,207.32,331.14,85.05,220.83,56.84,235.18,26.04,115.17,407.65,141.86,180.11,111.34,53.07  
2026-06-17,295.95,378.91,204.65,196.28,333.46,83.81,224.96,56.53,234.20,25.92,115.44,399.53,140.74,177.58,111.21,50.33  
2026-06-18,298.01,379.40,210.69,195.16,325.22,82.20,223.17,56.20,228.39,25.21,113.87,400.96,137.81,173.63,107.74,48.09  
2026-06-22,297.01,367.34,208.65,194.90,331.48,83.84,227.09,57.37,231.29,25.08,115.48,406.68,138.47,175.06,109.70,47.95  
2026-06-23,294.30,373.94,200.04,197.43,334.14,84.13,226.03,57.91,239.08,24.72,119.60,409.25,139.73,175.98,109.97,47.79  
2026-06-24,293.08,365.46,199.00,196.57,333.45,84.30,219.86,57.73,241.00,24.04,120.60,405.80,136.90,171.45,106.92,46.61  
2026-06-25,275.15,352.83,195.74,193.41,335.12,84.74,221.04,58.19,244.88,23.67,125.45,415.53,137.55,172.24,106.41,47.42  
2026-06-26,283.78,372.97,192.53,202.73,329.05,83.86,212.03,57.88,254.66,24.29,128.66,427.89,136.54,171.06,105.96,47.00  
2026-06-29,281.74,368.57,194.97,206.43,329.39,83.51,211.72,57.88,258.51,24.37,129.38,419.82,136.06,168.47,104.20,46.38  
2026-06-30,289.36,373.02,200.09,205.02,327.33,82.64,209.04,56.98,253.97,24.08,128.50,415.63,136.72,165.76,103.96,46.49  
2026-07-01,294.38,384.28,197.58,210.98,334.07,85.94,211.86,58.36,253.98,23.88,125.37,426.54,136.28,165.69,103.22,45.09  
2026-07-02,308.63,390.49,194.83,219.72,334.47,85.51,213.93,58.73,263.04,24.32,129.56,425.36,137.09,169.20,104.73,45.13  
2026-07-06,312.66,386.74,195.55,218.07,337.72,87.45,222.10,59.90,259.33,23.72,126.78,417.99,136.44,168.10,103.58,45.72  
2026-07-07,310.66,388.84,196.93,221.54,339.22,87.18,222.04,59.86,267.24,24.07,128.86,428.19,141.69,174.01,108.44,46.42  
2026-07-08,313.39,383.34,204.12,220.94,330.62,85.56,218.07,58.30,263.40,24.05,125.99,425.60,141.13,175.97,110.72,47.43  
2026-07-09,316.22,384.36,202.78,222.65,335.47,86.91,222.13,59.25,259.10,24.25,125.07,431.68,137.46,174.05,108.02,47.24  
2026-07-10,315.32,385.10,210.96,223.64,336.47,87.16,222.28,59.67,256.98,24.17,123.54,424.62,138.88,176.40,109.04,47.76  
2026-07-13,317.31,390.99,203.53,230.61,334.53,87.67,221.09,59.50,257.77,24.48,124.03,429.09,144.51,182.20,112.85,47.36  
2026-07-14,314.86,384.93,211.80,220.78,342.89,85.29,227.67,60.62,253.85,24.25,120.78,425.19,145.09,181.76,111.87,47.54  
2026-07-15,327.50,395.63,212.50,224.56,346.91,87.51,228.55,61.59,247.02,24.82,123.61,418.52,144.51,181.60,111.46,47.55  
2026-07-16,333.26,401.10,207.40,235.31,343.15,88.07,218.37,61.49,249.97,25.14,127.63,423.38,145.95,183.86,112.84,47.08  
2026-07-17,333.74,393.82,202.81,237.25,341.10,87.51,215.50,61.27,253.04,25.05,127.50,426.09,147.36,187.38,114.71,46.99  
2026-07-20,326.59,402.29,203.28,234.74,338.87,86.33,210.94,60.42,248.82,24.75,124.40,421.55,148.36,189.71,115.68,46.39  
2026-07-21,327.74,397.75,207.29,227.16,345.23,87.74,216.40,61.22,250.61,24.94,126.26,436.35,151.71,191.07,117.50,46.59  
2026-07-22,325.89,390.34,212.06,218.36,348.21,86.42,218.50,61.62,255.63,24.82,127.47,431.31,154.45,192.98,118.79,47.67  
2026-07-23,321.66,381.58,208.76,212.17,349.90,86.19,215.18,61.28,259.27,25.01,130.48,423.56,156.89,194.42,120.20,47.22  
2026-07-24,333.02,381.70,206.84,225.11,353.21,86.31,214.48,62.05,263.40,24.54,131.07,420.74,156.94,194.79,120.26,52.42  
2026-07-27,336.91,389.10,196.51,237.75,356.20,87.27,214.56,62.13,265.95,24.67,130.76,417.64,154.77,190.00,115.58,51.53  
2026-07-28,340.08,393.35,197.01,249.18,357.31,86.87,211.58,62.62,266.73,25.25,131.82,428.79,153.04,187.58,114.10,49.98  
2026-07-29,338.19,390.54,190.01,263.43,344.71,83.87,203.13,61.07,265.53,25.15,130.36,420.57,156.75,191.86,118.06,48.96  
2026-07-30,333.43,451.10,195.04,247.90,350.85,85.43,210.06,61.73,255.82,24.91,129.79,421.47,156.97,192.31,119.03,48.91  
2026-07-31,308.91,464.72,200.75,250.41,351.79,86.45,210.42,61.95,256.35,25.01,130.20,414.40,155.44,196.83,120.48,49.59  
2026-08-03,303.42,487.65,206.64,251.34,352.64,87.89,211.23,62.48,254.41,25.03,127.77,415.36,155.06,193.18,119.16,49.31  
2026-08-04,309.38,492.81,211.94,257.49,357.52,88.39,217.04,62.90,254.93,25.41,128.00,407.55,153.96,190.40,117.94,50.81  
2026-08-05,311.00,487.46,219.22,259.32,359.24,89.17,218.27,63.25,257.59,25.81,128.33,412.75,151.63,186.41,115.04,49.91  
2026-08-06,312.41,499.86,218.99,260.24,356.30,87.59,213.75,63.00,256.98,26.20,128.37,403.97,154.84,189.23,116.76,51.54  
2026-08-07,313.33,499.99,223.96,265.21,357.52,87.25,216.33,63.17,259.24,26.76,128.58,407.08,153.04,186.56,117.61,50.53